# GOVERNMENT OF MAHARASHTRA

OFFICIAL e-PROCUREMENT TENDER NOTIFICATION · Satellite Forest Canopy Density Monitoring & Wildlife Corridor Spatial Mapping

DETAILED TECHNICAL SPECIFICATIONS & SCOPE DOCUMENT

Tender Reference ID: 2026_MAH_5710685_34 Portal: https://tender.maharashtra.gov.in

Department / Authority: STATE FOREST & ENVIRONMENT DEPARTMENT

Published Date: 14 Oct 2026 | Online Bid Closing: 29 Oct 2026, 15:00 hrs

Tender Type: Open Competitive Global / National e-Bidding | Form of Contract: Turnkey Item Rate

EMD Value: 2% of Bid Value via e-Gateway | Tender Document Fee: INR 10,000 + 18% GST (Non-refundable)

## 1. EXECUTIVE SUMMARY & INVITATION TO BID

1.1 The Government of Maharashtra, acting through STATE FOREST & ENVIRONMENT DEPARTMENT, invites eligible and qualified agencies to submit technical and financial proposals online through the official state e-procurement portal for:

"Forest-Based Satellite-Based Forest Canopy Density & Wildlife Corridor Spatial Monitoring"

1.2 Brief Purpose: Multi-temporal satellite remote sensing imagery analysis, automated NDVI forest canopy density classification, wildlife migratory corridor tracking, forest boundary demarcation using DGPS pillars, and mobile GIS field patrol enforcement applications conforming to Forest Survey of India (FSI) standards.

1.3 Bidders are advised to thoroughly inspect the technical guidelines, bill of quantities (BOQ), minimum eligibility criteria, and general condition clauses before uploading their signed and sealed proposals.

## 2. CRITICAL DATES SHEET & PROCUREMENT TIMELINES

| Milestone Event Description | Official Schedule / Timestamp |
|---|---|
| Official Portal Publication Date | 14 Oct 2026 |
| Document Download Start Date | 14 Oct 2026, 10:00 hrs |
| Pre-Bid Query Submission Deadline | Within 7 Days of Portal Publication |
| Pre-Bid Clarification Meeting (Hybrid/VC) | 10th Day of Publication at 11:30 hrs |
| Bid Submission Start Date | 14 Oct 2026, 14:00 hrs |
| Online Bid Submission Due Date & Time | 29 Oct 2026, 15:00 hrs |
| Technical Bid Opening Date & Time | Next Working Day following Bid Closing at 15:30 hrs |
| Financial Bid (BOQ) Opening Date | To be notified online to technically qualified bidders |

## 3. COMPREHENSIVE TECHNICAL SCOPE & ENGINEERING METHODOLOGY

3.1 Overview of Engineering Operations:
The selected agency shall be responsible for turnkey end-to-end execution of all technical requirements described herein. All work must adhere strictly to National Geospatial Policy guidelines, Survey of India precision benchmarks, and relevant state standards.

3.2 Key Technical Components:
? Phase I ? Continuous Operating Reference Station (CORS) & DGPS Network: Establishment of primary Ground Control Points (GCPs) with Dual Frequency GNSS receivers linked to the state CORS network, achieving sub-2cm absolute spatial positioning accuracy.
? Phase II ? High-Precision Aerial / Terrestrial Sensor Acquisition: Deployment of calibrated multi-rotor/fixed-wing survey drones or Mobile Mapping Systems (MMS) equipped with high-resolution digital metric cameras and LiDAR sensors capable of delivering 5 cm Ground Sample Distance (GSD) orthophoto mosaics.
? Phase III ? Vector Extraction & Feature Classification: Digitization of building footprints, road alignments, parcel boundaries, and municipal infrastructure with structured attribute schemas conforming to national spatial data exchange standards.
? Phase IV ? Comprehensive Field Verification & Attribute Geo-tagging: Door-to-door verification using custom mobile GIS applications with automated GPS coordinates, photo capture, and offline cache synchronization.
? Phase V ? Enterprise Geodatabase & Decision Support Web GIS: Development and hosting of an enterprise PostgreSQL / PostGIS spatial database with GeoServer map services, OGC WMS/WFS interfaces, and role-based access control for department administrators.

### 3.3 Statutory Standards, Geodetic Accuracy & Delivery Format Requirements

? Coordinate Reference System: WGS84 Datum / UTM Projection (conforming to state UTM Zone).
? Horizontal Positional Accuracy: RMSE <= 5.0 cm on well-defined ground features.
? Vertical Precision: RMSE <= 10.0 cm for Digital Surface Model (DSM) and Digital Terrain Model (DTM).
? Orthomosaic Delivery: Uncompressed GeoTIFF and ECW format with complete pyramidal tiles.
? Vector Delivery: ESRI File Geodatabase (.gdb) and OGC GeoPackage (.gpkg) with topological validation.
? Attribute Data: Linked SQLite / PostgreSQL relational tables with audited integrity constraints.
? Cloud Deployment: High-availability tier with 99.9% uptime SLA and periodic daily automated snapshots.

### 3.4 Quality Control, Assurance & Third-Party Audit

The department reserves the right to appoint a Third-Party Quality Inspection Agency (TPIA) or Survey of India / State Remote Sensing Centre to validate 10% randomly selected sample plots across all deliverables. Any deliverable failing accuracy tolerances shall be resurveyed by the vendor at no additional cost to the department.

## 4. MINIMUM BIDDER QUALIFICATION & TECHNICAL ELIGIBILITY CRITERIA

Bidders must meet all of the following mandatory pre-qualification conditions as on the date of bid submission. Consortiums/Joint Ventures (JV) are permitted as per the conditions stated in Section 5.

| Criteria Category | Mandatory Qualification Threshold | Documentary Proof Required |
|---|---|---|
| Annual Financial Turnover | Average annual turnover of at least INR 12.00 Crores over the last 3 financial years (FY 2023-24, 2024-25, 2025-26). | Audited Balance Sheets & CA Certificate with UDIN |
| Past Experience | Completed at least two remote sensing forestry monitoring projects for State Forest Departments. | Client Completion Certificates & Work Orders |
| Quality Certifications | Valid ISO 9001:2015, ISO 27001 (Information Security), and minimum CMMI Level 3 (Services/Dev). | Valid Certificate Copies from Accredited Registrar |
| Legal Status | Registered Company under Indian Companies Act (or LLP) in operation for at least 5 years. | Certificate of Incorporation & GST Registration |
| Solvency Status | Bank Solvency Certificate for an amount not less than INR 4.80 Crores from a Scheduled Commercial Bank. | Original Bank Solvency Certificate in prescribed format |
| Debarment Undertaking | The bidder must not have been blacklisted or debarred by any Central/State Govt/PSU as on bid date. | Notarized Affidavit on INR 100 Stamp Paper |

### 4.2 Mandatory Key Technical Personnel to be Deployed on Project

| Designation / Role | Minimum Academic Qualification | Min Relevant Experience |
|---|---|---|
| Project Director / Team Leader | M.Tech / M.Sc Remote Sensing / GIS / Civil | 12+ Years Turnkey Project Leadership |
| Senior GIS Database Architect | B.Tech / MCA with PostgreSQL/PostGIS / Oracle | 8+ Years Spatial DB Architecture |
| Photogrammetry / LiDAR Specialist | M.Tech / B.Tech Geoinformatics / Photogrammetry | 8+ Years Aerial Sensor Processing |
| Field Survey & QA Lead Engineer | Degree/Diploma Civil + DGCA Drone Pilot Cert | 6+ Years DGPS & Ground Truthing |
| Full-Stack WebGIS Developer | B.Tech Computer Science / IT (React / Python) | 5+ Years Web Map Portal Dev |

## 5. PROJECT DELIVERABLE MILESTONES & PROGRESS PAYMENT SCHEDULE

| Milestone | Deliverable Description | Target Timeline | Payment (%) |
|---|---|---|---|
| M-1 | Inception Report, Project Execution Plan & Statutory Clearances | T0 + 30 Days | 10% |
| M-2 | Establishment of DGPS Ground Control Network & Raw Sensor Acquisition | T0 + 90 Days | 20% |
| M-3 | Ortho-mosaic Processing, Feature Extraction & 50% Field Verification | T0 + 180 Days | 25% |
| M-4 | 100% Field Survey Completion, UPIN Geo-tagging & Database Validation | T0 + 270 Days | 25% |
| M-5 | Web GIS Portal Deployment, User Acceptance Testing (UAT) & Training | T0 + 330 Days | 10% |
| M-6 | Final Handover, Final Documentation Sign-off & Commencement of AMC | T0 + 365 Days | 10% |

## 6. SERVICE LEVEL AGREEMENT (SLA), PENALTIES & PERFORMANCE GUARANTEE

? Performance Security (PBG): The successful bidder shall furnish a Performance Bank Guarantee equal to 5% of the total contract value within 14 days of the Letter of Award (LoA), valid until 60 days after completion of all contractual obligations.
? Delay Liquidated Damages: If the contractor fails to deliver milestones within agreed timelines, liquidated damages of 0.5% per week of delay (or fraction thereof) shall be deducted, subject to a maximum cap of 10% of total contract value.
? Defect Liability & Maintenance: 12 months comprehensive warranty from the date of final operational sign-off.
? Data Security & Ownership: All raw aerial imagery, point clouds, GIS geodatabases, and field registers shall be the sole and exclusive property of the Government of Maharashtra. The vendor shall not disclose or retain proprietary government data.

ISSUED FOR AND ON BEHALF OF:

Government of Maharashtra

STATE FOREST & ENVIRONMENT DEPARTMENT

[Digitally Signed & Validated on e-Procurement Portal]

Competent Tender Inviting Authority (TIA)